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Subject :- Advance Web Programming

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Practical:01

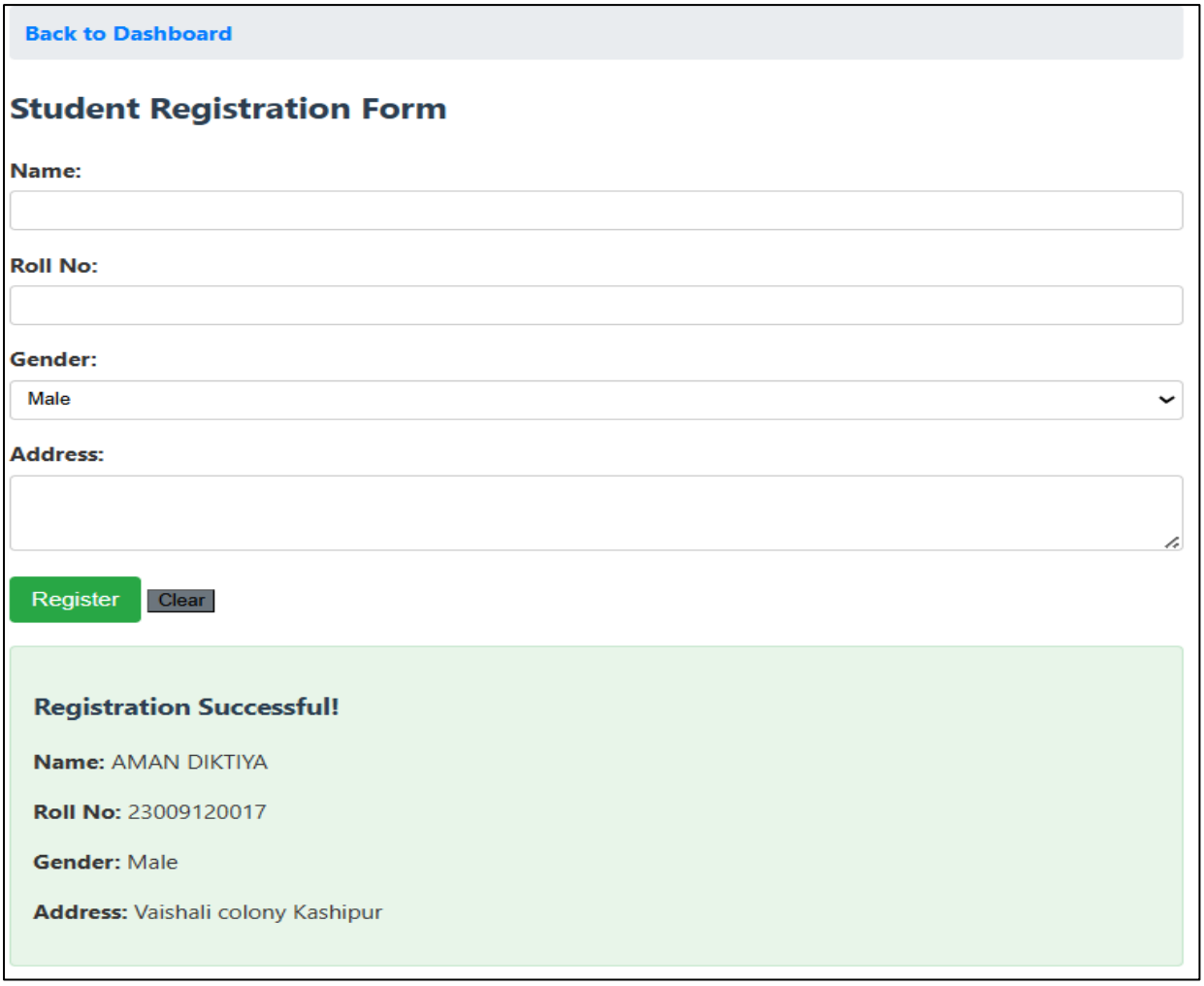
Aim: Write a program to create Student registration form.

Prac01.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 1: Student Registration Form</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
      <h2>Student Registration Form</h2>

      <form method="post" action="">
         <div class="form-group">
            <label>Name:</label>
            <input type="text" name="name" required>
         </div>
         <div class="form-group">
            <label>Roll No:</label>
            <input type="text" name="rollno" required>
         </div>
         <div class="form-group">
            <label>Gender:</label>
            <select name="gender">
               <option value="Male">Male</option>
               <option value="Female">Female</option>
            </select>
         </div>
         <div class="form-group">
            <label>Address:</label>
            <textarea name="address" rows="3"></textarea>
         </div>
         <button type="submit" name="submit">Register</button>
         <button type="reset" style="background-color: #6c757d;">Clear</button>
      </form>
      <?php
      if(isset($_POST['submit'])){
```

```
        $name = htmlspecialchars($_POST['name']);
        $rollno = htmlspecialchars($_POST['rollno']);
        $gender = $_POST['gender'];
        $address = htmlspecialchars($_POST['address']);
        echo "<div class='result-box'>";
        echo "<h3>Registration Successful!</h3>";
        echo "<p><strong>Name:</strong> $name</p>";
        echo "<p><strong>Roll No:</strong> $rollno</p>";
        echo "<p><strong>Gender:</strong> $gender</p>";
        echo "<p><strong>Address:</strong> $address</p>";
        echo "</div>";
    }
    ?>
  </div>
</body>
</html>
```

Output:-



Practical:02

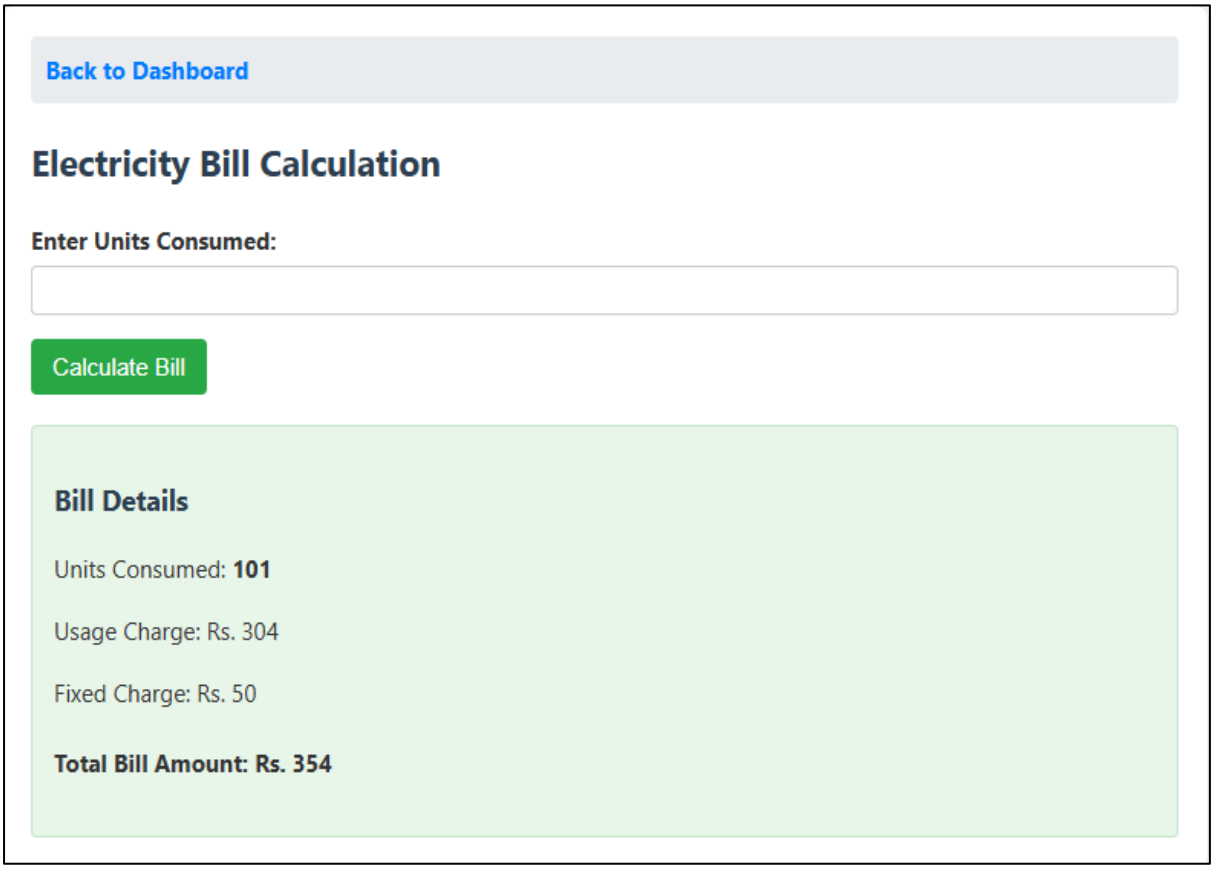
Aim: Write a program to perform EB bill calculation.

Prac02.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Ex 2: EB Bill Calculation</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="container">
        <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
        <h2>Electricity Bill Calculation</h2>

        <form method="post">
            <div class="form-group">
                <label>Enter Units Consumed:</label>
                <input type="number" name="units" required min="0">
            </div>
            <button type="submit" name="calculate">Calculate Bill</button>
        </form>

        <?php
        if(isset($_POST['calculate'])){
            $units = $_POST['units'];
            $bill = 0;

            // Logic:
            // First 100 units: Rs. 3/unit
            // Next 100 units (101-200): Rs. 4/unit
            // Above 200 units: Rs. 5/unit
            // Fixed charge: Rs. 50

            if($units <= 100) {
                $bill = $units * 3;
            } elseif($units <= 200) {
                $bill = (100 * 3) + (($units - 100) * 4);
            } else {
                $bill = (100 * 3) + (100 * 4) + (($units - 200) * 5);
```

```
        }

        $total_bill = $bill + 50; // Adding fixed charge

        echo "<div class='result-box'>";
        echo "<h3>Bill Details</h3>";
        echo "<p>Units Consumed: <strong>$units</strong></p>";
        echo "<p>Usage Charge: Rs. $bill</p>";
        echo "<p>Fixed Charge: Rs. 50</p>";
        echo "<h4>Total Bill Amount: Rs. $total_bill</h4>";
        echo "</div>";
    }
    ?>
  </div>
</body>
</html>
```

Output:-



Practical:03

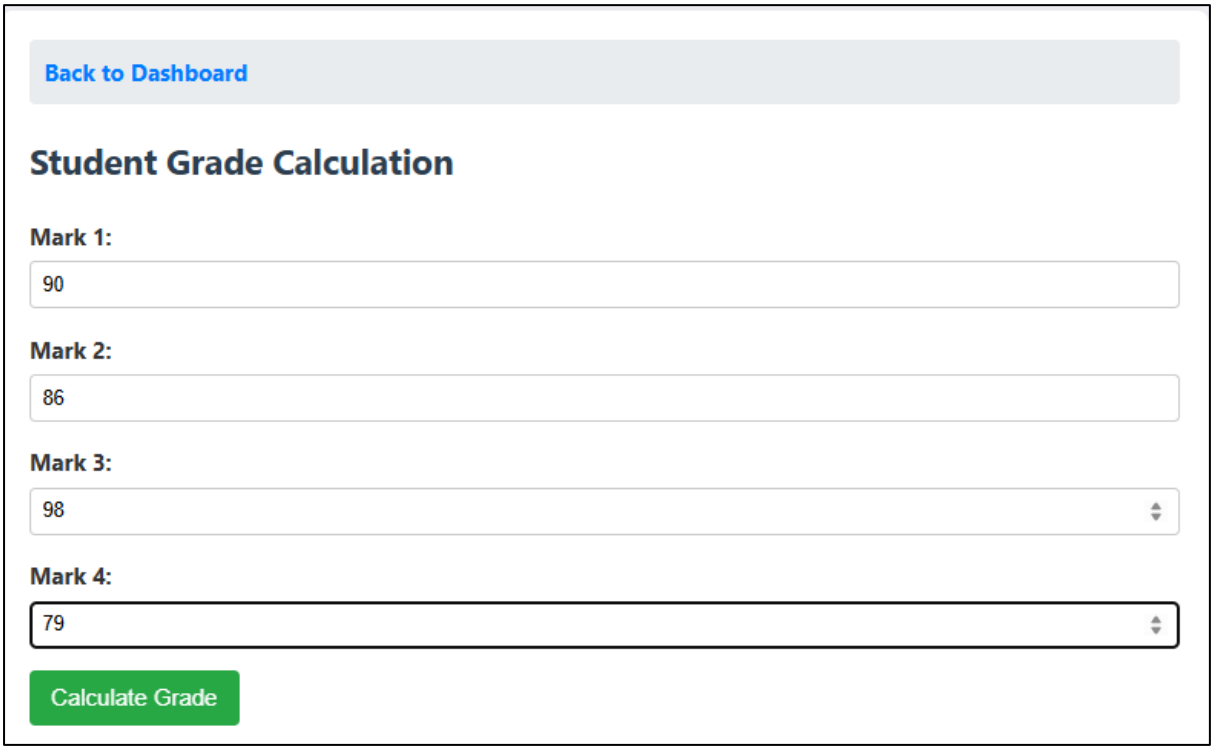
AIM : Write a program to perform Student grade manipulation.

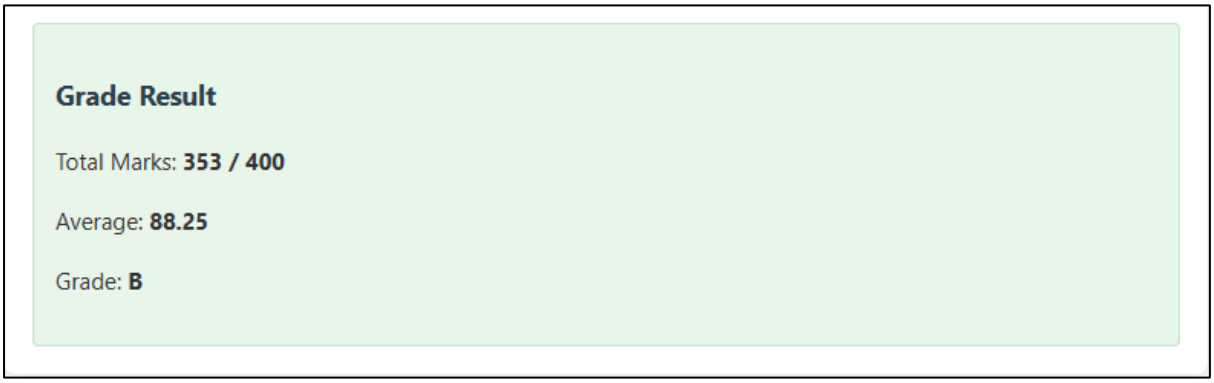
Prac03.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 3: Student Grade Manipulation</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>

   <div class="container">
      <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
      <h2>Student Grade Calculation</h2>
      <form method="post">
         <div class="form-group">
            <label>Mark 1:</label>
            <input type="number" name="m1" required min="0" max="100">
         </div>
         <div class="form-group">
            <label>Mark 2:</label>
            <input type="number" name="m2" required min="0" max="100">
         </div>
         <div class="form-group">
            <label>Mark 3:</label>
            <input type="number" name="m3" required min="0" max="100">
         </div>
         <div class="form-group">
            <label>Mark 4:</label>
            <input type="number" name="m4" required min="0" max="100">
         </div>
         <button type="submit" name="calc_grade">Calculate Grade</button>
      </form>

      <?php
      if(isset($_POST['calc_grade'])){
         $m1 = $_POST['m1'];
         $m2 = $_POST['m2'];
         $m3 = $_POST['m3'];
```

```
        $m4 = $_POST['m4'];

        $total = $m1 + $m2 + $m3 + $m4;
        $avg = $total / 4;
        $grade = "";

        if($m1 < 40 || $m2 < 40 || $m3 < 40 || $m4 < 40) {
           $grade = "Fail";
        } elseif($avg >= 90) {
           $grade = "A";
        } elseif($avg >= 80) {
           $grade = "B";
        } elseif($avg >= 70) {
           $grade = "C";
        } elseif($avg >= 60) {
           $grade = "D";
        } else {
           $grade = "E";
        }
        echo "<div class='result-box'>";
        echo "<h3>Grade Result</h3>";
        echo "<p>Total Marks: <strong>$total / 400</strong></p>";
        echo "<p>Average: <strong>$avg</strong></p>";
        echo "<p>Grade: <strong>$grade</strong></p>";
        echo "</div>";
     }
     ?>
   </div>
</body>
</html>
```

OUTPUT :


Back to Dashboard
Student Grade Calculation
Mark 1:
90
Mark 2:
86
Mark 3:
98
Mark 4:
79
Calculate Grade


Grade Result
Total Marks: 353 / 400
Average: 88.25
Grade: B

Practical:04

Aim - Write a program to perform String operations in PHP.

Prac04.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 4: String Operations</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
      <h2>String Operations in PHP</h2>

      <form method="post">
         <div class="form-group">
            <label>Enter a String:</label>
            <input type="text" name="string_input" required value="<?php echo
isset($_POST['string_input']) ? $_POST['string_input'] : ''; ?>">
         </div>
         <button type="submit" name="perform_ops">Perform Operations</button>
      </form>

      <?php
      if(isset($_POST['perform_ops'])){
         $str = $_POST['string_input'];

         echo "<div class='result-box'>";
         echo "<h3>Results for '$str':</h3>";
         echo "<ul>";
         echo "<li><strong>String Length (strlen):</strong> " . strlen($str) . "</li>";
         echo "<li><strong>Word Count (str_word_count):</strong> " . str_word_count($str) .
"</li>";
         echo "<li><strong>Reverse (strrev):</strong> " . strrev($str) . "</li>";
         echo "<li><strong>Uppercase (strtoupper):</strong> " . strtoupper($str) . "</li>";
         echo "<li><strong>Lowercase (strtolower):</strong> " . strtolower($str) . "</li>";
         echo "<li><strong>Title Case (ucwords):</strong> " . ucwords($str) . "</li>";
         echo "<li><strong>MD5 Hash (md5):</strong> " . md5($str) . "</li>";
         echo "<li><strong>Shuffle (str_shuffle):</strong> " . str_shuffle($str) . "</li>";
```

```
            echo "</ul>";
            echo "</div>";
        }
        ?>
    </div>
</body>
</html>
```

Output :

Back to Dashboard

String Operations in PHP

Enter a String:

45

Perform Operations

Results for '45':

- **String Length (strlen):** 2
- **Word Count (str_word_count):** 0
- **Reverse (strrev):** 54
- **Uppercase (strtoupper):** 45
- **Lowercase (strtolower):** 45
- **Title Case (ucwords):** 45
- **MD5 Hash (md5):** 6c8349cc7260ae62e3b1396831a8398f
- **Shuffle (str_shuffle):** 54

Practical:05

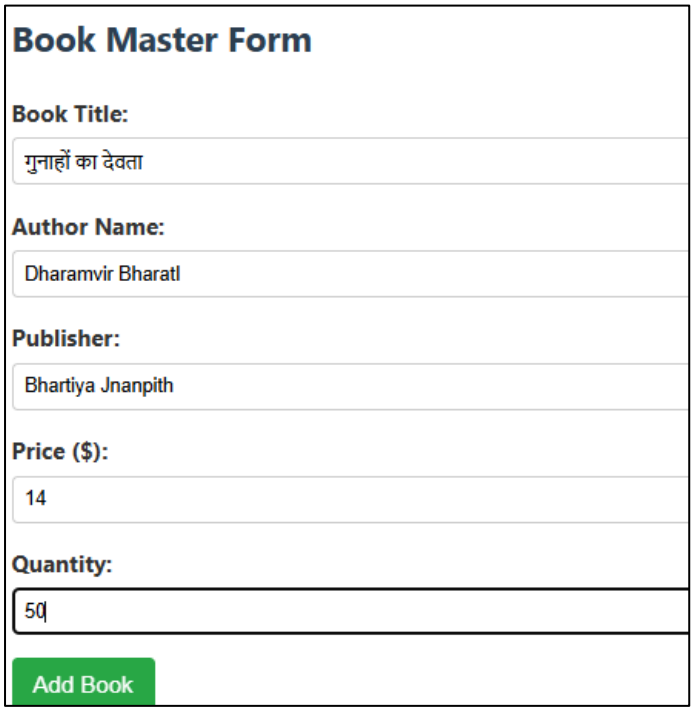
Aim - Write a program to create Book master form.

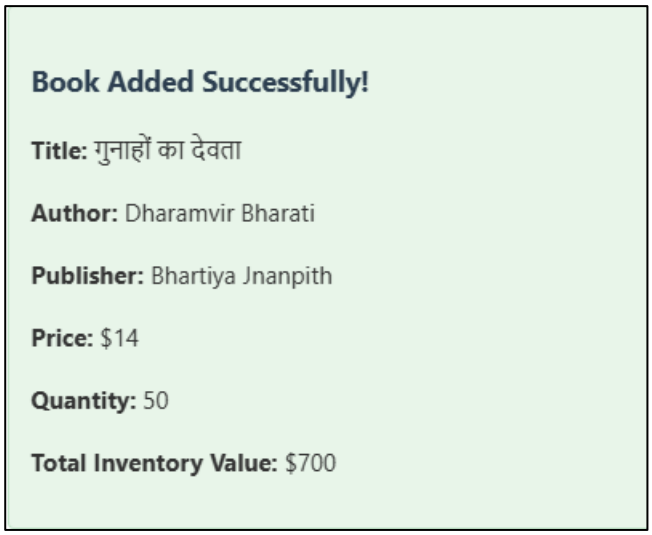
Prac05.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 5: Book Master Form</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
      <h2>Book Master Form</h2>

      <form method="post">
         <div class="form-group">
            <label>Book Title:</label>
            <input type="text" name="title" required>
         </div>
         <div class="form-group">
            <label>Author Name:</label>
            <input type="text" name="author" required>
         </div>
         <div class="form-group">
            <label>Publisher:</label>
            <input type="text" name="publisher" required>
         </div>
         <div class="form-group">
            <label>Price ($):</label>
            <input type="number" name="price" step="0.01" required>
         </div>
         <div class="form-group">
            <label>Quantity:</label>
            <input type="number" name="qty" required>
         </div>
         <button type="submit" name="submit_book">Add Book</button>
      </form>

      <?php
```

```
        if(isset($_POST['submit_book'])){
            $title = htmlspecialchars($_POST['title']);
            $author = htmlspecialchars($_POST['author']);
            $publisher = htmlspecialchars($_POST['publisher']);
            $price = $_POST['price'];
            $qty = $_POST['qty'];

            $total_cost = $price * $qty;

            echo "<div class='result-box'>";
            echo "<h3>Book Added Successfully!</h3>";
            echo "<p><strong>Title:</strong> $title</p>";
            echo "<p><strong>Author:</strong> $author</p>";
            echo "<p><strong>Publisher:</strong> $publisher</p>";
            echo "<p><strong>Price:</strong> $$price</p>";
            echo "<p><strong>Quantity:</strong> $qty</p>";
            echo "<p><strong>Total Inventory Value:</strong> $$total_cost</p>";
            echo "</div>";
        }
        ?>
    </div>
</body>
</html>
```

Output :





Practical:06

Aim - Write a program to perform Form validation – Railway ticket reservation.

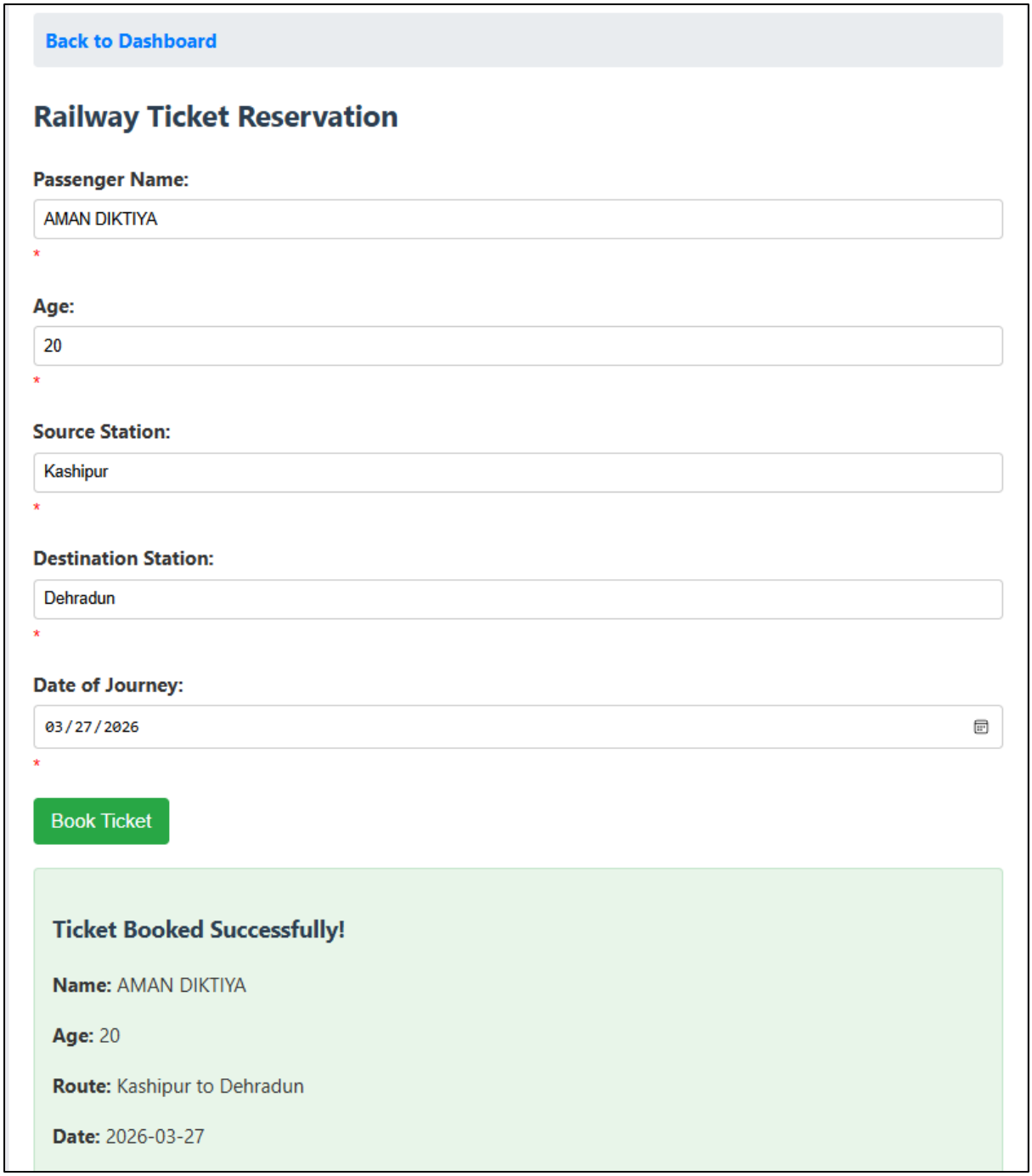
Prac06.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Ex 6: Railway Ticket Reservation</title>
    <link rel="stylesheet" href="style.css">
    <style>
        .error { color: red; font-size: 0.9em; }
    </style>
</head>
<body>
    <div class="container">
        <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
        <h2>Railway Ticket Reservation</h2>

        <?php
        $nameErr = $ageErr = $sourceErr = $destErr = $dateErr = "";
        $name = $age = $source = $dest = $date = "";
        $isValid = true;

        if ($_SERVER["REQUEST_METHOD"] == "POST") {
            if (empty($_POST["name"])) {
                $nameErr = "Name is required";
                $isValid = false;
            } else {
                $name = htmlspecialchars($_POST["name"]);
                if (!preg_match("/^[a-zA-Z-' ]*$/",$name)) {
                    $nameErr = "Only letters and white space allowed";
                    $isValid = false;
                }
            }

            if (empty($_POST["age"])) {
                $ageErr = "Age is required";
                $isValid = false;
            } else {
```

```
            $age = $_POST["age"];
            if ($age < 5 || $age > 100) {
                $ageErr = "Age must be between 5 and 100";
                $isValid = false;
            }
        }

        if (empty($_POST["source"])) {
            $sourceErr = "Source is required";
            $isValid = false;
        } else {
            $source = htmlspecialchars($_POST["source"]);
        }

        if (empty($_POST["dest"])) {
            $destErr = "Destination is required";
            $isValid = false;
        } else {
            $dest = htmlspecialchars($_POST["dest"]);
            if (strtolower($source) == strtolower($dest)) {
                $destErr = "Source and Destination cannot be same";
                $isValid = false;
            }
        }

        if (empty($_POST["date"])) {
            $dateErr = "Date is required";
            $isValid = false;
        } else {
            $date = $_POST["date"];
            if ($date < date("Y-m-d")) {
                $dateErr = "Date cannot be in the past";
                $isValid = false;
            }
        }
    }
    ?>

    <form method="post" action="<?php echo
htmlspecialchars($_SERVER["PHP_SELF"]);?>">
        <div class="form-group">
            <label>Passenger Name:</label>
            <input type="text" name="name" value="<?php echo $name;?>">
            <span class="error">* <?php echo $nameErr;?></span>
```

```
        </div>
        <div class="form-group">
            <label>Age:</label>
            <input type="number" name="age" value="<?php echo $age;?>">
            <span class="error">* <?php echo $ageErr;?></span>
        </div>
        <div class="form-group">
            <label>Source Station:</label>
            <input type="text" name="source" value="<?php echo $source;?>">
            <span class="error">* <?php echo $sourceErr;?></span>
        </div>
        <div class="form-group">
            <label>Destination Station:</label>
            <input type="text" name="dest" value="<?php echo $dest;?>">
            <span class="error">* <?php echo $destErr;?></span>
        </div>
        <div class="form-group">
            <label>Date of Journey:</label>
            <input type="date" name="date" value="<?php echo $date;?>">
            <span class="error">* <?php echo $dateErr;?></span>
        </div>
        <button type="submit">Book Ticket</button>
    </form>

    <?php
    if ($_SERVER["REQUEST_METHOD"] == "POST" && $isValid) {
        echo "<div class='result-box'>";
        echo "<h3>Ticket Booked Successfully!</h3>";
        echo "<p><strong>Name:</strong> $name</p>";
        echo "<p><strong>Age:</strong> $age</p>";
        echo "<p><strong>Route:</strong> $source to $dest</p>";
        echo "<p><strong>Date:</strong> $date</p>";
        echo "</div>";
    }
    ?>
  </div>
</body>
</html>
```

Output :


Back to Dashboard
Railway Ticket Reservation
Passenger Name:
AMAN DIKTIYA
Age:
20
Source Station:
Kashipur
Destination Station:
Dehradun
Date of Journey:
03/27/2026
Book Ticket
Ticket Booked Successfully!
Name: AMAN DIKTIYA
Age: 20
Route: Kashipur to Dehradun
Date: 2026-03-27

Practical:07

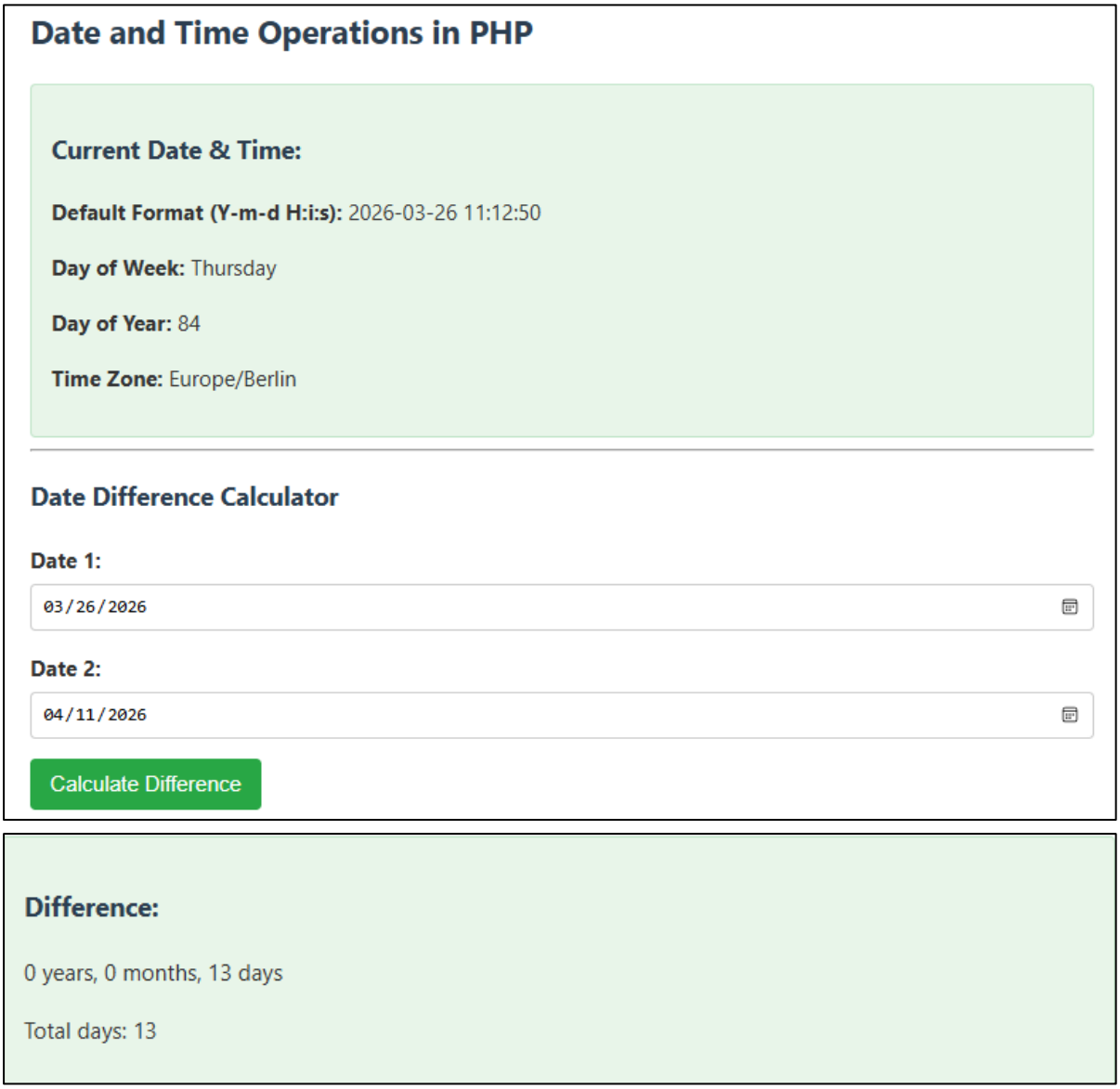
Aim - Write a program to perform Date and time operations in PHP.

Prac07.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 7: Date and Time Operations</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
      <h2>Date and Time Operations in PHP</h2>
      <div class="result-box">
         <h3>Current Date & Time:</h3>
         <p><strong>Default Format (Y-m-d H:i:s):</strong> <?php echo date("Y-m-d H:i:s");
?></p>
         <p><strong>Day of Week:</strong> <?php echo date("l"); ?></p>
         <p><strong>Day of Year:</strong> <?php echo date("z"); ?></p>
         <p><strong>Time Zone:</strong> <?php echo date_default_timezone_get(); ?></p>
      </div>
      <hr>
      <h3>Date Difference Calculator</h3>
      <form method="post">
         <div class="form-group">
            <label>Date 1:</label>
            <input type="date" name="date1" required>
         </div>
         <div class="form-group">
            <label>Date 2:</label>
            <input type="date" name="date2" required>
         </div>
         <button type="submit" name="calc_diff">Calculate Difference</button>
      </form>

      <?php
      if(isset($_POST['calc_diff'])){
         $d1 = new DateTime($_POST['date1']);
```

```
        $d2 = new DateTime($_POST['date2']);
        $interval = $d1->diff($d2);
        echo "<div class='result-box'>";
        echo "<h3>Difference:</h3>";
        echo "<p>" . $interval->y . " years, " . $interval->m . " months, " . $interval->d . "
days</p>";
        echo "<p>Total days: " . $interval->days . "</p>";
        echo "</div>";
    }
    ?>
  </div>
</body>
</html>
```

Output :-



Practical:08

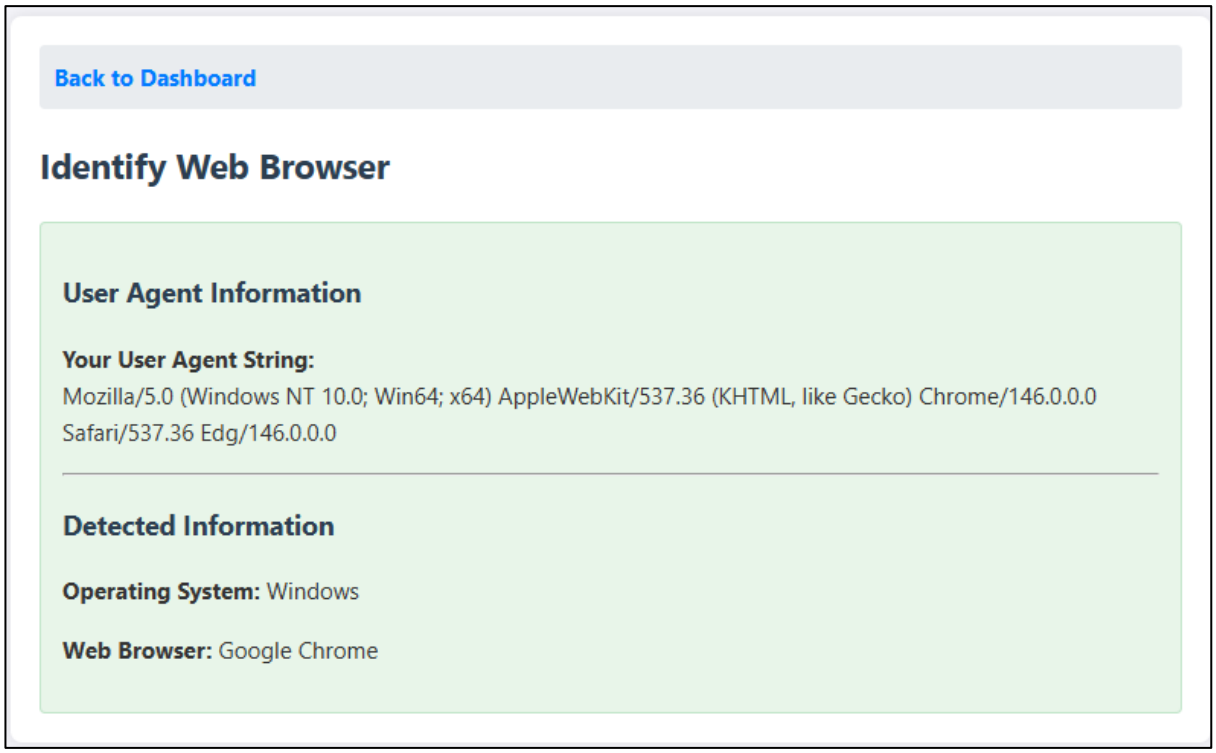
Aim- Write a program to Identify the web browser.

Prac08.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 8: Identify Web Browser</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links"><a href="index.php">Back to Dashboard</a></div>
      <h2>Identify Web Browser</h2>

      <?php
      $agent = $_SERVER['HTTP_USER_AGENT'];
      $browser = "Unknown";
      $platform = "Unknown";

      // Simple detection logic
      if (preg_match('/linux/i', $agent)) {
         $platform = 'Linux';
      } elseif (preg_match('/macintosh|mac os x/i', $agent)) {
         $platform = 'Mac';
      } elseif (preg_match('/windows|win32/i', $agent)) {
         $platform = 'Windows';
      }

      if(preg_match('/MSIE/i',$agent) && !preg_match('/Opera/i',$agent)) {
         $browser = 'Internet Explorer';
      } elseif(preg_match('/Firefox/i',$agent)) {
         $browser = 'Mozilla Firefox';
      } elseif(preg_match('/Chrome/i',$agent)) {
         $browser = 'Google Chrome';
      } elseif(preg_match('/Safari/i',$agent)) {
         $browser = 'Apple Safari';
      } elseif(preg_match('/Opera/i',$agent)) {
         $browser = 'Opera';
      } elseif(preg_match('/Netscape/i',$agent)) {
```

```
        $browser = 'Netscape';
    }
    ?>

    <div class="result-box">
        <h3>User Agent Information</h3>
        <p><strong>Your User Agent String:</strong><br> <?php echo $agent; ?></p>
        <hr>
        <h3>Detected Information</h3>
        <p><strong>Operating System:</strong> <?php echo $platform; ?></p>
        <p><strong>Web Browser:</strong> <?php echo $browser; ?></p>
    </div>
  </div>
</body>
</html>
```

Output :



Practical:09

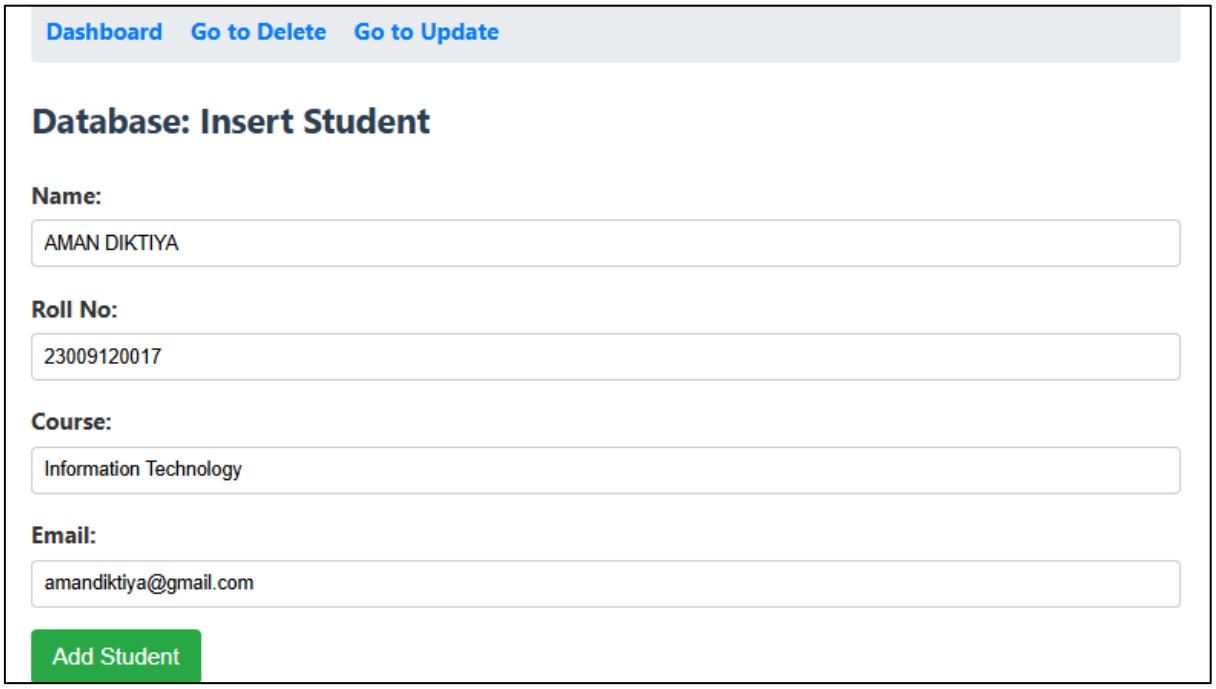
Aim- Demonstrate the Database – Insert operation.

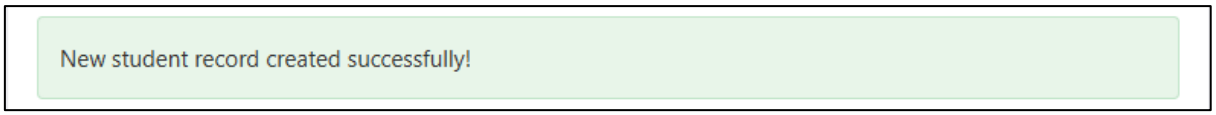
Prac09.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 9: Database Insert</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links">
         <a href="index.php">Dashboard</a>
         <a href=" Prac10.php">Go to Delete</a>
         <a href=" Prac11.php">Go to Update</a>
      </div>
      <h2>Database: Insert Student</h2>

      <form method="post">
         <div class="form-group">
            <label>Name:</label>
            <input type="text" name="name" required>
         </div>
         <div class="form-group">
            <label>Roll No:</label>
            <input type="text" name="roll_no" required>
         </div>
         <div class="form-group">
            <label>Course:</label>
            <input type="text" name="course" required>
         </div>
         <div class="form-group">
            <label>Email:</label>
            <input type="email" name="email" required>
         </div>
         <button type="submit" name="insert">Add Student</button>
      </form>

      <?php
```

```
    require_once 'db_connect.php';

    if(isset($_POST['insert'])){
      try {
        $sql = "INSERT INTO students (name, roll_no, course, email) VALUES (:name,
:roll_no, :course, :email)";
        $stmt = $pdo->prepare($sql);
        $stmt->bindParam(':name', $_POST['name']);
        $stmt->bindParam(':roll_no', $_POST['roll_no']);
        $stmt->bindParam(':course', $_POST['course']);
        $stmt->bindParam(':email', $_POST['email']);

        $stmt->execute()
        echo "<div class='result-box'>New student record created successfully!</div>";
      } catch(PDOException $e) {
        echo "<div class='error-box'>Error: " . $e->getMessage() . "</div>";
      }
    }
    ?>
  </div>
</body>
</html>
```

Output :





Practical:10

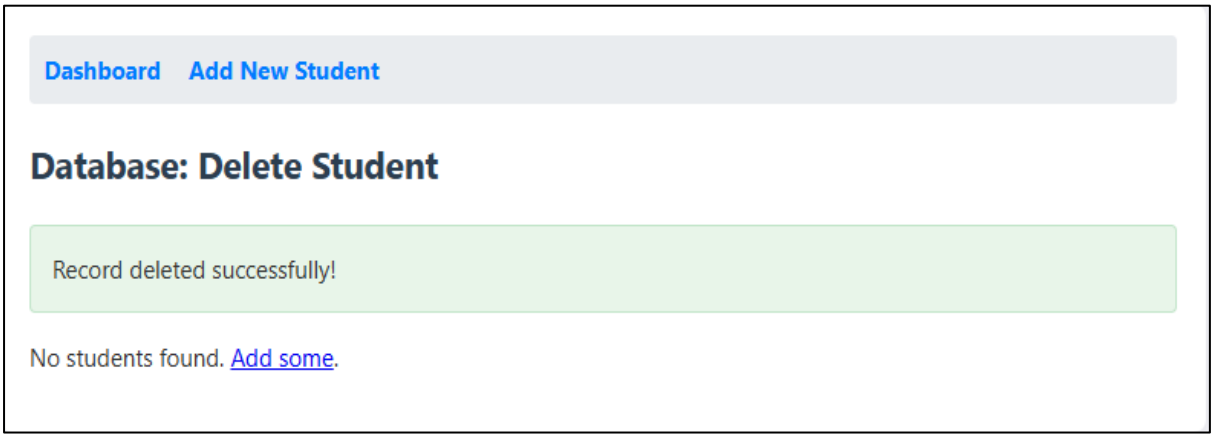
Aim- Demonstrate the Database – Delete operation.

Prac10.php:-

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 10: Database Delete</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links">
         <a href="index.php">Dashboard</a>
         <a href=" Prac09.php">Add New Student</a>
      </div>
      <h2>Database: Delete Student</h2>

      <?php
      require_once 'db_connect.php';

      // Handle Delete
      if(isset($_POST['delete_id'])){
         try {
            $sql = "DELETE FROM students WHERE id = :id";
            $stmt = $pdo->prepare($sql);
            $stmt->bindParam(':id', $_POST['delete_id']);
            $stmt->execute();
            echo "<div class='result-box'>Record deleted successfully!</div>";
         } catch(PDOException $e) {
            echo "<div class='error-box'>Error deleting: " . $e->getMessage() . "</div>";
         }
      }

      // List Students
      try {
         $stmt = $pdo->query("SELECT * FROM students");
         $students = $stmt->fetchAll(PDO::FETCH_ASSOC);

         if(count($students) > 0) {
```

```
            echo "<table>";
            echo "<tr><th>ID</th><th>Name</th><th>Roll
No</th><th>Course</th><th>Action</th></tr>";
            foreach($students as $row) {
               echo "<tr>";
               echo "<td>" . $row['id'] . "</td>";
               echo "<td>" . $row['name'] . "</td>";
               echo "<td>" . $row['roll_no'] . "</td>";
               echo "<td>" . $row['course'] . "</td>";
               echo "<td>
                     <form method='post' style='margin:0;' onsubmit='return confirm(\"Are you
sure?\");'>
                        <input type='hidden' name='delete_id' value='" . $row['id'] . "'>
                        <button type='submit' style='background-color: #dc3545; padding: 5px
10px; font-size: 14px;'>Delete</button>
                     </form>
                    </td>";
               echo "</tr>";
            }
            echo "</table>";
         } else {
            echo "<p>No students found. <a href=' Prac09.phpphp'>Add some</a>.</p>";
         }
      } catch(PDOException $e) {
         echo "<div class='error-box'>Error fetching data: " . $e->getMessage() . "</div>";
      }
      ?>
   </div>
</body>
</html>
```

Output :



Practical:11

Aim- Demonstrate the Database - Update operation.

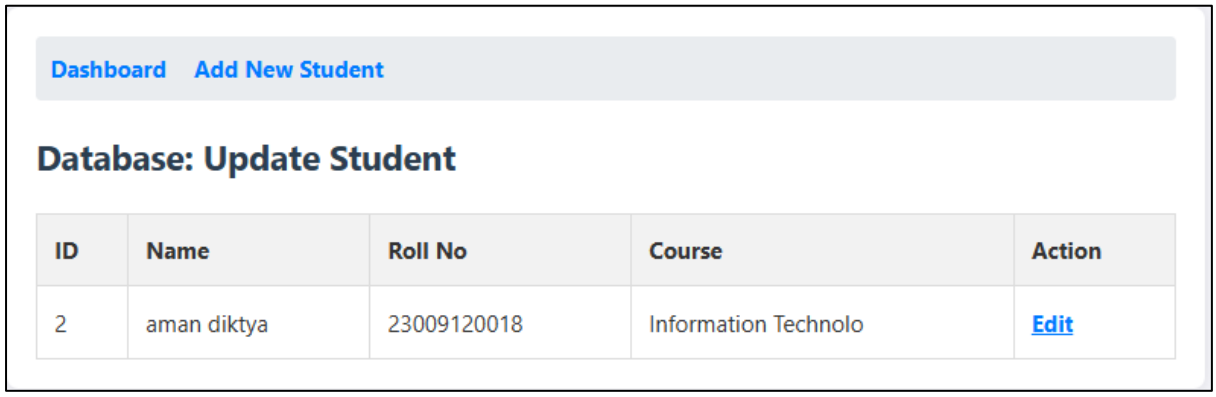
Prac11.php:-

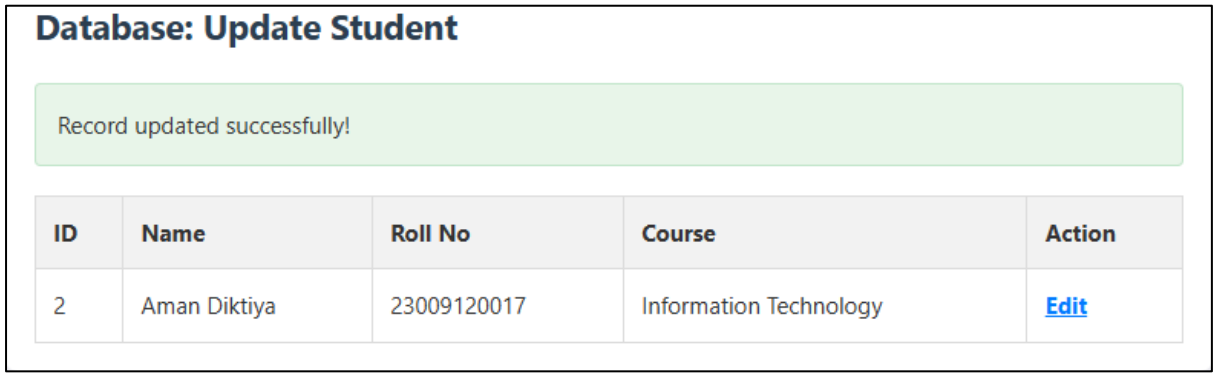
```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Ex 11: Database Update</title>
   <link rel="stylesheet" href="style.css">
</head>
<body>
   <div class="container">
      <div class="nav-links">
         <a href="index.php">Dashboard</a>
         <a href="Prac09.php">Add New Student</a>
      </div>
      <h2>Database: Update Student</h2>

      <?php
      require_once 'db_connect.php';

      // 2. Handle Update Submission
      if(isset($_POST['update_record'])){
         try {
            $sql = "UPDATE students SET name=:name, roll_no=:roll_no, course=:course,
email=:email WHERE id=:id";
            $stmt = $pdo->prepare($sql);
            $stmt->execute([
               ':name' => $_POST['name'],
               ':roll_no' => $_POST['roll_no'],
               ':course' => $_POST['course'],
               ':email' => $_POST['email'],
               ':id' => $_POST['id']
            ]);
            echo "<div class='result-box'>Record updated successfully!</div>";
         } catch(PDOException $e) {
            echo "<div class='error-box'>Error updating: " . $e->getMessage() . "</div>";
         }
      }

      // 1. Show Edit Form if requested
      if(isset($_GET['edit_id'])){
         $edit_id = $_GET['edit_id'];
```

```
        $stmt = $pdo->prepare("SELECT * FROM students WHERE id = :id");
        $stmt->execute([':id' => $edit_id]);
        $student = $stmt->fetch(PDO::FETCH_ASSOC);

        if($student):
    ?>
        <div class="result-box" style="background-color: #e3f2fd; border-color: #90caf9;">
            <h3>Edit Record</h3>
            <form method="post" action="Prac11.php" style="margin-top:0;">
                <input type="hidden" name="id" value="<?php echo $student['id']; ?>">
                <div class="form-group">
                    <label>Name:</label>
                    <input type="text" name="name" value="<?php echo $student['name']; ?>"
required>
                </div>
                <div class="form-group">
                    <label>Roll No:</label>
                    <input type="text" name="roll_no" value="<?php echo $student['roll_no'];
?>" required>
                </div>
                <div class="form-group">
                    <label>Course:</label>
                    <input type="text" name="course" value="<?php echo $student['course']; ?>"
required>
                </div>
                <div class="form-group">
                    <label>Email:</label>
                    <input type="email" name="email" value="<?php echo $student['email']; ?>"
required>
                </div>
                <button type="submit" name="update_record">Update Student</button>
                <a href="Prac11.php" style="margin-left: 10px;">Cancel</a>
            </form>
        </div>
        <hr>
    <?php
        endif;
    }

    // 3. List Students
    try {
        $stmt = $pdo->query("SELECT * FROM students");
        $students = $stmt->fetchAll(PDO::FETCH_ASSOC);

        if(count($students) > 0) {
            echo "<table>";
```

```
            echo "<tr><th>ID</th><th>Name</th><th>Roll
No</th><th>Course</th><th>Action</th></tr>";
            foreach($students as $row) {
                echo "<tr>";
                echo "<td>" . $row['id'] . "</td>";
                echo "<td>" . $row['name'] . "</td>";
                echo "<td>" . $row['roll_no'] . "</td>";
                echo "<td>" . $row['course'] . "</td>";
                echo "<td><a href='Prac11.php?edit_id=" . $row['id'] . "' style='color: #007bff;
font-weight: bold;'>Edit</a></td>";
                echo "</tr>";
            }
            echo "</table>";
        } else {
            echo "<p>No students found. <a href='Prac09.php'>Add some</a>.</p>";
        }
    } catch(PDOException $e) {
        echo "<div class='error-box'>Error fetching data: " . $e->getMessage() . "</div>";
    }
    ?>
  </div>
</body>
</html>
```

Output :





Practical:12

Aim- Demonstrate the Queries Record selection operation.

Prac12.sql:-

```
CREATE DATABASE college_db;
USE college_db;

CREATE TABLE students (
   rollno INT PRIMARY KEY,
   name VARCHAR(50),
   branch VARCHAR(20),
   marks INT
);
INSERT INTO students VALUES
(101,'Aman','CSE',85),
(102,'Ravi','ECE',72),
(103,'Neha','CSE',90),
(104,'Karan','ME',65);

SELECT * FROM students;
```

Search Results | Export

rollno int(11)	name varchar(50)	branch varchar(20)	marks int(11)
101	Aman	CSE	85
102	Ravi	ECE	72
103	Neha	CSE	90
104	Karan	ME	65

```
SELECT name, branch FROM students;
```

Search Results

name varchar(50)	branch varchar(20)
Aman	CSE
Ravi	ECE
Neha	CSE
Karan	ME

SELECT * FROM students
WHERE branch = 'CSE';

rollno int(11)	name varchar(50)	branch varchar(20)	marks int(11)
101	Aman	CSE	85
103	Neha	CSE	90

SELECT * FROM students
WHERE branch='CSE' AND marks>80;

SELECT * FROM students
WHERE branch='CSE' OR branch='ECE';

rollno int(11)	name varchar(50)	branch varchar(20)	marks int(11)
101	Aman	CSE	85
102	Ravi	ECE	72
103	Neha	CSE	90

SELECT * FROM students
WHERE name LIKE 'A%';

rollno int(11)	name varchar(50)	branch varchar(20)	marks int(11)
101	Aman	CSE	85

SELECT * FROM students
ORDER BY marks DESC;

rollno int(11)	name varchar(50)	branch varchar(20)	marks int(11)
103	Neha	CSE	90
101	Aman	CSE	85
102	Ravi	ECE	72
104	Karan	ME	65

Practical:13

Aim- Write the queries to demonstrate the working with date and time functions.

Prac13.sql:-

```
CREATE DATABASE Poly_db;

USE Poly_db;

CREATE TABLE students (
   rollno INT PRIMARY KEY,
   name VARCHAR(50),
   admission_date DATE,
   admission_time TIME,
   created_at DATETIME
);

INSERT INTO students VALUES
(101,'Aman','2023-07-10','10:30:00','2023-07-10 10:30:00'),
(102,'Ravi','2023-07-12','11:15:00','2023-07-12 11:15:00'),
(103,'Neha','2023-07-15','09:45:00','2023-07-15 09:45:00');

SELECT CURDATE();

SELECT CURTIME();

SELECT NOW();

SELECT name, YEAR(admission_date) AS Year
FROM students;

SELECT name, MONTH(admission_date) AS Month
FROM students;

SELECT name, DAY(admission_date) AS Day
FROM students;

SELECT name, DATE(created_at) AS Only_Date
FROM students;
```

Cost: 2ms 1

CURDATE() date
2026-03-26 00:00:00

Cost: 3ms

CURTIME() time
16:57:40

Cost: 4ms 1

NOW() datetime
2026-03-26 16:59:43

name varchar(50)	Year
Aman	2023
Ravi	2023
Neha	2023

name varchar(50)	Month
Aman	7
Ravi	7
Neha	7

name varchar(50)	Day
Aman	10
Ravi	12
Neha	15

name varchar(50)	Only_Date
Aman	2023-07-10 00:00:00
Ravi	2023-07-12 00:00:00
Neha	2023-07-15 00:00:00

SELECT name, DATEDIFF(CURDATE(), admission_date) AS Days_Difference
FROM students;

Cost: 24ms 1 Total 3

name varchar(50)	Days_Difference
Aman	990
Ravi	988
Neha	985

SELECT name, DATE_ADD(admission_date, INTERVAL 10 DAY) AS New_Date
FROM students;

Cost: 10ms 1 Total 3

name varchar(50)	New_Date
Aman	2023-07-20 00:00:00
Ravi	2023-07-22 00:00:00
Neha	2023-07-25 00:00:00

SELECT name, DATE_SUB(admission_date, INTERVAL 5 DAY) AS Old_Date
FROM students;

Cost: 6ms 1 Total 3

name varchar(50)	Old_Date
Aman	2023-07-05 00:00:00
Ravi	2023-07-07 00:00:00
Neha	2023-07-10 00:00:00

SELECT * FROM students
WHERE admission_date BETWEEN '2023-07-10' AND '2023-07-12';

students

Search Results Export Cost: 12ms 1 Total 2

rollno int(11)	name varchar(50)	admission_date date	admission_time time	created_at datetime
101	Aman	2023-07-10 00:00:00	10:30:00	2023-07-10 10:30:00
102	Ravi	2023-07-12 00:00:00	11:15:00	2023-07-12 11:15:00

Practical:14

Aim- Demonstrate the Simple Programming and File handling operation

 # Simple Program to add two numbers:-

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
sum = a + b
print("Sum is:", sum)
```

```
PS C:\xampp\htdocs\AWP\Prac14> python -u "c:\xampp\htdocs\AWP\Prac14\1Simple Programming Example.py"
Enter first number: 3
Enter second number: 4
Sum is: 7
```

#File create or write:-

```
file = open("student.txt", "w")
file.write("Name: Aman Diktiya\nBranch: IT\nMarks: 85")
file.close()
```

```
Prac14 > student.txt
1  Name: Aman Diktiya
2  Branch: IT
3  Marks: 85
```

#File Append data add :-

```
file = open("student.txt", "a")
file.write("\nGrade: A")
file.close()
```

```
Prac14 > student.txt
1  Name: Aman Diktiya
2  Branch: IT
3  Marks: 85
4  Grade: A
```

#File read:-

```
file = open("student.txt", "r")
content = file.read()
print(content)
file.close()
```

```
PS C:\xampp\htdocs\AWP\Prac14> python -u "c:\xampp\htdocs\AWP\Prac14\4File Read.py"
Name: Aman Diktiya
Branch: IT
Marks: 85
Grade: A
```